

L^AT_EX Reference Sheet

A. J. A. & Claude

July 14, 2026

Contents

1	Text Formatting	3
1.1	The Basics	3
1.2	Live Examples	3
1.3	Coloured Statement Line	3
1.4	Typewriter and Verbatim Text	3
2	Fonts	4
2.1	The Three Built-In Families	4
2.2	Changing the Whole Document's Font	5
2.3	Making Sans-Serif the Default	5
2.4	System Fonts (XeLaTeX / LuaLaTeX only)	5
2.5	Font Sizes	5
2.6	A Warning for Template-Based Papers	6
3	Lists	6
3.1	Nested Numbered List	6
3.2	Bulleted and Description Lists	6
4	Equations	6
4.1	Numbered vs. Unnumbered	6
4.2	Piecewise Functions (<code>cases</code>)	7
4.3	The Normal Distribution	7
4.4	Aligning Equations (<code>align</code>)	7
4.5	Splitting Big Equations	7
4.6	The Leap Year Formula	8
5	Tables	8
5.1	Multirow and Multicolumn	8
5.2	Empty Cells	8
5.3	Gotchas	9
5.4	Colouring Table Backgrounds	9
6	Figures	11
6.1	Single Figure	11
6.2	Subfigures Side by Side	11
6.3	Subfigures Stacked Vertically	11

6.4	Referencing	11
6.5	Keeping Floats in Their Section	12
7	Algorithms	12
8	Flowcharts with TikZ	12
8.1	Step 1: Preamble Setup	13
8.2	Step 2: Placing Nodes	13
8.3	Step 3: Connecting with Arrows	13
8.4	Full Working Example	14
8.5	Quick Reference	15
9	Sections, Contents, Footnotes	15
9.1	Unnumbered Sections	15
9.2	Table of Contents	15
9.3	Footnotes	15
10	Symbols	15
10.1	Greek Letters	16
10.2	Relations	16
10.3	Arithmetic and Sets	16
10.4	Arrows	16
10.5	Calculus and Analysis	16
10.6	Logic and Miscellaneous	16
10.7	Text-Mode Symbols (no math mode)	16
10.8	Reserved Characters (Escaping)	17
11	References	18
11.1	Manual — thebibliography	18
11.2	BibTeX — .bib file	18
12	Compile Checklist	18

1 Text Formatting

1.1 The Basics

Effect	Command	Package
Bold	<code>\textbf{text}</code>	—
Italic	<code>\textit{text}</code>	—
Underline	<code>\underline{text}</code>	—
Emphasis	<code>\emph{text}</code>	—
Strikethrough	<code>\sout{text}</code>	ulem
Small caps	<code>\textsc{text}</code>	—
Monospace	<code>\texttt{text}</code>	—
Sans-serif	<code>\textsf{text}</code>	—
Slanted	<code>\textsl{text}</code>	—
Superscript	<code>text</code>	—
Subscript	<code>\textsubscript{text}</code>	—
Uppercase	<code>\MakeUppercase{text}</code>	—
Text colour	<code>\textcolor{red}{text}</code>	xcolor
Highlight	<code>\colorbox{yellow}{text}</code>	xcolor

1.2 Live Examples

Bold text. *Italic text.* Underlined text. SMALL CAPS TEXT. Monospace text. Sans-serif text. Slanted text.

Red text. Blue text. Custom hex colour. Highlighted text.

Strikethrough: it's around ~~365~~ 365.24 days.

Nested: **Bold and blue at once.**

Font sizes: small normal large Large LARGE huge

1.3 Coloured Statement Line

Leap Year = **The full equation is shown on the screen/projector.**

```
\noindent Leap Year = \textbf{\textcolor{blue}{The full equation ... }}
```

1.4 Typewriter and Verbatim Text

Two different tools that look similar but behave differently.

`\texttt` — **Font Change Only**

Switches to a monospace font. $\text{\LaTeX}$ still processes commands and special characters inside it, so you can nest other formatting:

- INSERT THE EQUATION SHOWN IN THE PROJECTOR
- Bold typewriter
- Blue typewriter

- **Bold, blue, typewriter**

```
\texttt{INSERT THE EQUATION SHOWN IN THE PROJECTOR}
\texttt{\textbf{\textcolor{blue}{Bold, blue, typewriter}}}
```

`\verb` and `verbatim` — No Processing at All

Renders text literally. Special characters like `\`, `$`, `%`, `_` are printed as-is instead of being interpreted — which is exactly what you need when showing L^AT_EX source code (every code block in this document uses it).

Inline: `\frac{a}{b}` costs 100% \$\$ — printed raw, nothing interpreted.

```
\verb|\frac{a}{b} costs 100% $$|
```

Block form:

```
\begin{verbatim}
  This is printed exactly as typed.
  Even \commands and $math$ and % comments survive.
```

└ (closing tag)

Which to Use

You want	Use
Monospace look, still want bold/colour	<code>\texttt{}</code>
Show literal code / special characters	<code>\verb</code> or <code>verbatim</code>
A command name in a sentence	<code>\texttt{}</code>
A multi-line code listing	<code>verbatim</code> environment

Notes on `\verb`: the delimiter can be any character not appearing in the text — `\verb|...|`, `\verb!...!`, and `\verb+...+` are all valid. It cannot be used inside another command’s argument (so no `\verb` inside `\caption{}` or `\textbf{}`) — use `\texttt` there instead.

2 Fonts

2.1 The Three Built-In Families

No package needed — L^AT_EX always has these:

- Roman / serif (the default) — `\textrm{}` or `{\rmfamily ...}`
- Sans-serif — `\textsf{}` or `{\sffamily ...}`
- Monospace / typewriter — `\texttt{}` or `{\ttfamily ...}`

The `\text..{}` form takes an argument; the `\..family` form switches until the end of the current group.

2.2 Changing the Whole Document's Font

Load one of these in the preamble:

```
\usepackage{times}      % Times Roman
\usepackage{helvet}     % Helvetica (sans)
\usepackage{courier}    % Courier (mono)
\usepackage{palatino}   % Palatino
\usepackage{bookman}    % Bookman
\usepackage{charter}    % Charter
\usepackage{lmodern}    % Latin Modern -- an improved default
```

The modern, better-maintained versions — these also fix the math font to match, which the older packages do not:

```
\usepackage[T1]{fontenc}      % better encoding -- add alongside
\usepackage{newtxtext,newtxmath} % Times + matching math
\usepackage{mathptmx}         % Times (older, still common)
```

2.3 Making Sans-Serif the Default

```
\renewcommand{\familydefault}{\sfdefault}
```

Goes in the preamble, after any font packages. Use `\ttdefault` for monospace throughout (rarely a good idea for a whole document).

2.4 System Fonts (XeLaTeX / LuaLaTeX only)

```
\usepackage{fontspec}
\setmainfont{Georgia}
\setsansfont{Arial}
\setmonofont{Consolas}
```

The most flexible option — any font installed on the system, including OpenType. **But** it requires compiling with XeLaTeX or LuaLaTeX instead of pdfL^AT_EX. On Overleaf: Menu → Compiler. It will not work under plain pdfL^AT_EX, so it is a real commitment, not a drop-in.

2.5 Font Sizes

Independent of the family. These switch until the end of the group:

tiny scriptsize footnotesize small normalsize
large Large LARGE huge Huge

```
{\tiny ...} {\scriptsize ...} {\footnotesize ...} {\small ...}
{\normalsize ...} {\large ...} {\Large ...} {\LARGE ...}
{\huge ...} {\Huge ...}
```

The document's base size is a class option — everything above scales relative to it:

```
\documentclass[12pt]{article} % 10pt (default), 11pt, or 12pt
```

2.6 A Warning for Template-Based Papers

Journal and conference classes — `acmart`, `IEEE`, `Springer` — specify their typeface deliberately. **Do not override it.** Changing the document font will make the output stop matching the required format, which on a reproduction assignment defeats the purpose. The family switches in the first subsection (`\texttt`, `\textsf`) are fine for local emphasis; it is the document-wide changes that cause trouble.

3 Lists

3.1 Nested Numbered List

1. A year is a leap year if:
 - (a) It is divisible by 4 and not divisible by 100
 - (b) Or it is divisible by 400.
2. A year is NOT a leap year in all other cases.

```
\begin{enumerate}
  \item A year is a leap year if:
    \begin{enumerate}[label=(\alph*)]
      \item Divisible by 4 and not by 100
      \item Or divisible by 400.
    \end{enumerate}
  \item NOT a leap year otherwise.
\end{enumerate}
```

3.2 Bulleted and Description Lists

- First bullet
- Second bullet

Julian Introduced 45 BC.

Gregorian Introduced 1582.

The `enumitem` package customises labels: `[label=(\alph*)]`, `[label=\roman*.]`, `[label=\arabic*)]`.

4 Equations

4.1 Numbered vs. Unnumbered

Numbered — use `equation`:

$$E = mc^2 \tag{1}$$

Unnumbered — add a star, `equation*`:

$$E = mc^2$$

4.2 Piecewise Functions (cases)

$$|y| = \begin{cases} y & \text{if } y \geq 0 \\ -y & \text{if } y < 0 \end{cases}$$

$$p(t) = \begin{cases} 0, & t < 0, \\ e^{-t}, & 0 \leq t < 3, \\ \ln(t-2), & t \geq 3 \end{cases}$$

```
\begin{equation*}
|y| =
\begin{cases}
y & \text{if } y \geq 0 \\
-y & \text{if } y < 0
\end{cases}
\end{equation*}
```

4.3 The Normal Distribution

$$\mathcal{N}(x_i, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

4.4 Aligning Equations (align)

Put `&` immediately before the character to align on (usually `=`), and `\\` to end each line.

$$\begin{aligned} T &= \sum_{i=1}^n (x_i - \mu)^2 \\ &= \sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2 \end{aligned}$$

```
\begin{align*}
T &= \sum_{i=1}^n (x_i - \mu)^2 \\
&= \sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2
\end{align*}
```

4.5 Splitting Big Equations

`split` — one equation, several lines, one number:

$$\begin{aligned} f(x) &= x^4 + 4x^3 - 3x^2 \\ &\quad + 2x - 7 + \frac{1}{x^2 + 1} \end{aligned} \tag{2}$$

`multline` — first line flush left, last flush right:

$$a + b + c + d + e + f + g + h + i + j$$

$$+ k + l + m + n + o + p + q + r + s + t$$

Which to use:

Situation	Environment
One equation, 2+ lines, numbered once	<code>split in equation</code>
One long equation, no natural break	<code>multline</code>
Several steps, each numbered	<code>align</code>
Auto line-breaking	<code>dmath (breqn)</code>

4.6 The Leap Year Formula

$$\text{Leap Year}(Y) = \begin{cases} \text{True,} & \text{if } (Y \bmod 4 = 0 \text{ and } Y \bmod 100 \neq 0) \\ & \text{or } (Y \bmod 400 = 0) \\ \text{False,} & \text{otherwise} \end{cases} \quad (3)$$

5 Tables

5.1 Multirow and Multicolumn

`\multicolumn{n}{align}{content}` merges horizontally. `\multirow{n}{width}{content}` merges vertically (needs `\usepackage{multirow}`). Use `*` for the width to mean “natural width”.

Year	Divisible By			Leap Year?
	4?	100?	400?	
1900	Yes	Yes	No	No
2000	Yes	Yes	Yes	Yes
2024	Yes	No	No	Yes

Table 1: Leap Year Calculation Examples

```
\begin{tabular}{|c|c|c|c|c|}
\hline
\multirow{2}{*}{Year} & \multicolumn{3}{c|}{Divisible By}
& \multirow{2}{*}{Leap Year?} \\
& 4? & 100? & 400? & \\
\cline{2-4}
& 4? & 100? & 400? & \\
\hline
1900 & Yes & Yes & No & No \\
\hline
\end{tabular}
```

% <-- empty cells hold the multirow slots

5.2 Empty Cells

Leave nothing between the `&` separators. You must still supply an empty `&` for every column a `\multirow` from a previous row is occupying — $\text{\LaTeX}$ will not skip it for you. For “no data”, prefer a visible placeholder:

A	B	C
1		3
1	—	3
1	N/A	3

Table 2: Truly empty vs. explicit “no data”

5.3 Gotchas

- $\wedge$ and $_$ are **math-mode only**. $0(n^{\{2\}})$ in a table cell errors out — write $\$0(n^{\{2\}})\$$. Math mode already italicises, so `\textit{}` is redundant.
- `\multicolumn{1}{*}{...}` is invalid — the alignment argument must be `l`, `c`, or `r` (optionally with `l`). `*` is only valid as the width argument of `\multirow`.
- Every row needs `\\` before the next `\hline`.
- `\cline{2-4}` draws a partial rule under columns 2–4 only — use it instead of `\hline` when a `\multirow` spans across that boundary.

5.4 Colouring Table Backgrounds

Requires `\usepackage[table]{xcolor}` — the `[table]` option is essential; plain `xcolor` will not work.

Single Cell

Use `\cellcolor{colour}` at the start of the cell.

A	B	C
1	2	3
4	5	6

Table 3: Individual cell colouring

```
1 & \cellcolor{yellow} 2 & 3 \\
4 & 5 & \cellcolor{green!30} 6 \\
```

Whole Row

Use `\rowcolor{colour}` at the start of the row.

Year	Divisible by 4?	Leap Year?
1900	Yes	No
2000	Yes	Yes
2024	Yes	Yes

Table 4: Row colouring — often used for headers

```
\rowcolor{gray!30}
Year & Divisible by 4? & Leap Year? \\
```

Whole Column

Put `>\columncolor{colour}` in the column specification. Needs `\usepackage{array}`.

Year	Divisible by 4?	Leap Year?
1900	Yes	No
2024	Yes	Yes

Table 5: Column colouring

```
\begin{tabular}{|>\columncolor{yellow!25}}c|c|c|}
```

Alternating Rows (Zebra Striping)

`\rowcolors{start}{oddc colour}{even colour}` — note the plural `rowcolors`. Place it immediately before `\begin{tabular}`.

Year	Divisible by 4?	Leap Year?
1900	Yes	No
2000	Yes	Yes
2024	Yes	Yes
1800	No	No

Table 6: Zebra striping, starting from row 2

```
\rowcolors{2}{gray!15}{white} % start at row 2, so header stays uncoloured
\begin{tabular}{|c|c|c|}
```

Defining Custom Colours

```
\definecolor{lightblue}{RGB}{173,216,230}
\definecolor{mygreen}{HTML}{90EE90}
```

```
\rowcolor{lightblue}
\cellcolor{mygreen}
```

Colour Intensity

The `!` syntax mixes a colour with white: `blue!20` is 20% blue, 80% white (a pale tint); `blue!100` (or just `blue`) is full strength. Values in the 15–30 range work well for table backgrounds — anything darker tends to fight with the text.

Precedence

When several rules apply to the same cell, the more specific one wins:

`\cellcolor > \rowcolor > \rowcolors > \columncolor`

So a `\cellcolor` will override the row's colour, and a `\rowcolor` overrides the column's.

6 Figures

6.1 Single Figure

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.6\textwidth]{image.jpg}
  \caption{Description of the figure}
  \label{fig:myimage}
\end{figure}
```

6.2 Subfigures Side by Side

`\hfill` between the blocks is what puts them on one row. The outer caption numbers the figure; the inner captions become (a), (b), ...

```
\usepackage{subcaption}

\begin{figure}[h]
  \centering
  \begin{subfigure}[b]{0.4\textwidth}
    \centering
    \includegraphics[width=\textwidth]{julius.jpg}
    \caption{Julius}
    \label{fig:julius}
  \end{subfigure}
  \hfill % <-- side by side
  \begin{subfigure}[b]{0.4\textwidth}
    \centering
    \includegraphics[width=\textwidth]{gregory.jpg}
    \caption{Gregory}
    \label{fig:gregory}
  \end{subfigure}
  \caption{Pioneers of Modern Calendars}
  \label{fig:pioneers}
\end{figure}
```

6.3 Subfigures Stacked Vertically

Identical, but **drop the** `\hfill` and leave a blank line between the two `subfigure` blocks instead.

6.4 Referencing

As shown in Figure~\ref{fig:pioneers}, both rulers reformed the calendar.

Notes: `\label` must come after `\caption`. The `~` is a non-breaking space so “Figure” never separates from its number. Compile **twice** for references to resolve.

6.5 Keeping Floats in Their Section

L^AT_EX floats do not respect section boundaries by default. Fixes:

- `\usepackage[section]{placeins}` — automatic barrier at every `\section`. Best default.
- `\FloatBarrier` — manual barrier, from `placeins`.
- `[H]` — from `\usepackage{float}`, forces exact placement. Note the capital H.

7 Algorithms

Requires `\usepackage{algorithm}` and `\usepackage{algpseudocode}`. The `[1]` turns on line numbering.

```
\begin{algorithm}
\caption{Check if a Year is a Leap Year}
\label{alg:leapyear}
\begin{algorithmic}[1]
  \Procedure{IsLeapYear}{$Y$}
    \If{$Y \bmod 4 = 0$ \textbf{and} $Y \bmod 100 \neq 0$}
      \State \Return True
    \ElsIf{$Y \bmod 400 = 0$}
      \State \Return True
    \Else
      \State \Return False
    \EndIf
  \EndProcedure
\end{algorithmic}
\end{algorithm}
```

Common commands:

Construct	Command
Statement	<code>\State ...</code>
If / else if / else	<code>\If{} \ElsIf{} \Else \EndIf</code>
For loop	<code>\For{} ... \EndFor</code>
While loop	<code>\While{} ... \EndWhile</code>
Procedure	<code>\Procedure{N}{args} \EndProcedure</code>
Return	<code>\State \Return value</code>
Comment	<code>\Comment{text}</code>

Reference it with `Algorithm~\ref{alg:leapyear}`.

8 Flowcharts with TikZ

Flowcharts are drawn with TikZ. The idea has three parts: (1) define shape styles once in the preamble, (2) place nodes using those styles, (3) connect them with arrows. Build up slowly — these rarely come out right on the first compile.

8.1 Step 1: Preamble Setup

```
\usepackage{tikz}
\usetikzlibrary{shapes.geometric, arrows.meta, positioning}
```

Then define a style for each kind of box you'll need. A style is just a named bundle of options you reuse:

```
\tikzstyle{startstop} = [rectangle, rounded corners,
    minimum width=2.5cm, minimum height=0.9cm,
    text centered, draw=black, fill=red!30]
```

```
\tikzstyle{process} = [rectangle,
    minimum width=2.5cm, minimum height=0.9cm,
    text centered, draw=black, fill=blue!20]
```

```
\tikzstyle{decision} = [diamond, aspect=2,
    minimum width=2.5cm, minimum height=0.9cm,
    text centered, draw=black, fill=green!20]
```

```
\tikzstyle{arrow} = [thick, -{Stealth[length=3mm]}]
```

What each piece means: `rectangle` / `diamond` is the shape; `rounded corners` softens a rectangle into a start/stop box; `fill=red!30` is the background (30% red); `draw=black` is the border; `-{Stealth}` makes a proper arrowhead.

8.2 Step 2: Placing Nodes

Each node has the form:

```
\node (name) [style, position] {visible text};
```

(name) is an internal handle you'll use to draw arrows — it is not printed. [style] is one of the styles above. {text} is what appears in the box.

Positioning is relative to a node already placed:

- `below of=start` — put this node below the one named `start`
- `right of=dec2` — put it to the right of `dec2`
- `node distance=1.8cm` (set once on the picture) controls the default gap
- `yshift=-0.8cm` / `xshift=3cm` — nudge further when the default spacing overlaps (diamonds are tall, so decision nodes often need an extra nudge)

8.3 Step 3: Connecting with Arrows

```
\draw [arrow] (start) -- (input); % plain arrow
\draw [arrow] (dec1) -- node[anchor=west] {Yes} (dec2); % labelled arrow
```

(a) -- (b) draws a straight line from node `a` to node `b`. To label the arrow (the “Yes”/“No” on decision branches), drop a `node{...}` in the middle; `anchor=west` just decides which side of the line the label sits on (`west` = left, `east` = right, `south` = below).

8.4 Full Working Example

Putting all three steps together (styles are in the preamble):

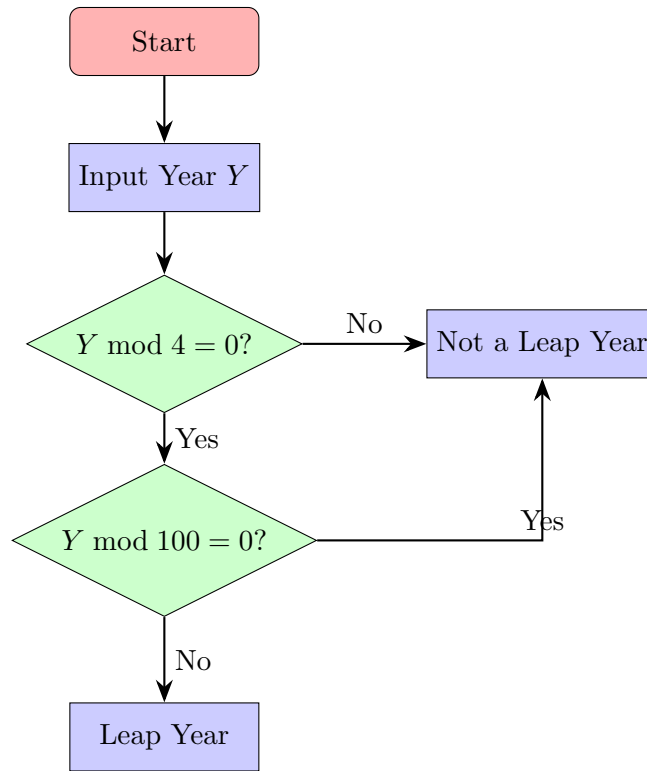


Figure 1: Leap Year Determination Flowchart

```
\begin{tikzpicture}[node distance=1.8cm]
\node (start) [startstop] {Start};
\node (input) [process, below of=start] {Input Year  $Y$ };
\node (dec1) [decision, below of=input, yshift=-0.4cm] { $Y \bmod 4 = 0?$ };
\node (dec2) [decision, below of=dec1, yshift=-0.8cm] { $Y \bmod 100 = 0?$ };
\node (leap) [process, below of=dec2, yshift=-0.8cm] {Leap Year};
\node (notleap) [process, right of=dec1, xshift=3.2cm] {Not a Leap Year};

\draw [arrow] (start) -- (input);
\draw [arrow] (input) -- (dec1);
\draw [arrow] (dec1) -- node[anchor=south] {No} (notleap);
\draw [arrow] (dec1) -- node[anchor=west] {Yes} (dec2);
\draw [arrow] (dec2) -- node[anchor=west] {No} (leap);
\draw [arrow] (dec2.east) --| node[anchor=south] {Yes} (notleap);
\end{tikzpicture}
```

8.5 Quick Reference

Want	Do
Start/stop box	<code>rectangle, rounded corners</code>
Process box	<code>rectangle</code>
Decision box	<code>diamond</code>
Straight arrow	<code>(a) -- (b)</code>
Right-angle arrow	<code>(a) - (b)</code> or <code>(a) - (b)</code>
Label an arrow	<code>-- node{Yes} (b)</code>
Move a node down	<code>below of=X, yshift=-...</code>
Fix overlaps	increase <code>yshift</code> / <code>xshift</code>

Table 7: TikZ flowchart quick reference

One new thing in this example: `-|` (instead of `--`) draws a right-angled arrow — horizontal first, then vertical. Handy for routing a “Yes” branch around the side of the chart instead of straight through another box. `|-` does the reverse (vertical first, then horizontal).

9 Sections, Contents, Footnotes

9.1 Unnumbered Sections

Add a star: `\section*{Disclaimer}`. This also removes it from the table of contents. To put it back:

```
\section*{Disclaimer}
\addcontentsline{toc}{section}{Disclaimer}
```

9.2 Table of Contents

```
\tableofcontents          % place after \maketitle
\setcounter{tocdepth}{2}  % 1=sections, 2+=subsections, 3+=subsubsections
```

Compile **twice** — the first pass writes the `.toc` file, the second reads it back.

9.3 Footnotes

A footnote here.¹ No package needed — just `\footnote{...}` at the point where the marker should appear.

```
...evaluating students' \LaTeX\footnote{A document preparation system.}
proficiency.
```

Note `\LaTeX` swallows the following space — write `\LaTeX{}` when a space must follow.

10 Symbols

All of these need math mode ($\dots$) unless marked otherwise.

¹This is the footnote text; the number is generated automatically.

10.1 Greek Letters

α	<code>\alpha</code>	β	<code>\beta</code>	γ	<code>\gamma</code>
Γ	<code>\Gamma</code>	δ	<code>\delta</code>	Δ	<code>\Delta</code>
ϵ	<code>\epsilon</code>	ε	<code>\varepsilon</code>	θ	<code>\theta</code>
λ	<code>\lambda</code>	μ	<code>\mu</code>	π	<code>\pi</code>
σ	<code>\sigma</code>	Σ	<code>\Sigma</code>	ϕ	<code>\phi</code>
φ	<code>\varphi</code>	ω	<code>\omega</code>	Ω	<code>\Omega</code>

10.2 Relations

$\neq$	<code>\neq</code>	$\leq$	<code>\leq</code>	$\geq$	<code>\geq</code>
$\approx$	<code>\approx</code>	$\equiv$	<code>\equiv</code>	$\propto$	<code>\propto</code>
$\ll$	<code>\ll</code>	$\gg$	<code>\gg</code>	$\sim$	<code>\sim</code>

10.3 Arithmetic and Sets

$\times$	<code>\times</code>	$\div$	<code>\div</code>	$\pm$	<code>\pm</code>
$\mp$	<code>\mp</code>	$\cdot$	<code>\cdot</code>	$\in$	<code>\in</code>
$\notin$	<code>\notin</code>	$\subset$	<code>\subset</code>	$\subseteq$	<code>\subseteq</code>
$\cup$	<code>\cup</code>	$\cap$	<code>\cap</code>	$\emptyset$	<code>\emptyset</code>
$\forall$	<code>\forall</code>	$\exists$	<code>\exists</code>	$\neg$	<code>\neg</code>

10.4 Arrows

$\rightarrow$	<code>\rightarrow</code>	$\leftarrow$	<code>\leftarrow</code>	$\Rightarrow$	<code>\Rightarrow</code>
$\Leftarrow$	<code>\Leftarrow</code>	$\leftrightarrow$	<code>\leftrightarrow</code>	$\Leftrightarrow$	<code>\Leftrightarrow</code>
$\mapsto$	<code>\mapsto</code>	$\rightarrow$	<code>\rightarrow</code>	$\uparrow$	<code>\uparrow</code>

10.5 Calculus and Analysis

$\int$	<code>\int</code>	$\iint$	<code>\iint</code>	$\oint$	<code>\oint</code>
$\sum$	<code>\sum</code>	$\prod$	<code>\prod</code>	∂	<code>\partial</code>
∇	<code>\nabla</code>	∞	<code>\infty</code>	$\sqrt{x}$	<code>\sqrt{x}</code>
$\sqrt[n]{x}$	<code>\sqrt[n]{x}</code>	$\frac{a}{b}$	<code>\frac{a}{b}</code>	$\lim$	<code>\lim</code>

10.6 Logic and Miscellaneous

$\wedge$	<code>\wedge</code>	$\vee$	<code>\vee</code>	$\oplus$	<code>\oplus</code>
90°	<code>90^\circ</code>	$\circ$	<code>\circ</code>	$\dots$	<code>\dots</code>
$\lceil x \rceil$	<code>\lceil x \rceil</code>	$\lfloor x \rfloor$	<code>\lfloor x \rfloor</code>	$\lvert x \rvert$	<code>\lvert x \rvert</code>
$a \bmod n$	<code>a \bmod n</code>	$\star$	<code>\star</code>	$\dagger$	<code>\dagger</code>

10.7 Text-Mode Symbols (no math mode)

$--$	<code>--</code>	$-$	<code>-</code>	$\S$	<code>\S</code>
$\copyright$	<code>\copyright</code>	$\%$	<code>\%</code>	$\&$	<code>\&</code>
$\#$	<code>\#</code>	$-$	<code>-</code>	$"x"$	<code>"x"</code>

Note on mod: `\bmod` is the binary operator ($a \bmod n$, spaced like +). `\pmod` is for congruences: $a \equiv b \pmod{n}$.

10.8 Reserved Characters (Escaping)

Ten characters have special meaning in $\text{\LaTeX}$. Typing them raw either breaks the compile or does something unintended. Each has an escape form.

Char	Type this	Reserved because
&	<code>\&</code>	Column separator in tables
%	<code>\%</code>	Starts a comment
\$	<code>\\$</code>	Toggles math mode
#	<code>\#</code>	Macro parameter
-	<code>_</code>	Subscript (math mode)
{	<code>\{</code>	Groups content
}	<code>\}</code>	Groups content
~	<code>\textasciitilde</code>	Non-breaking space
^	<code>\textasciicircum</code>	Superscript (math mode)
\	<code>\textbackslash</code>	Starts every command

Table 8: The ten reserved characters

The Three That Break the Pattern

The first seven take a plain backslash. The last three do not:

- **Backslash** — you cannot write `\\`, because that means line break. Use `\textbackslash`.
- **Caret** — `\^` produces an accent ($\hat{o}$), not a literal caret. Use `\textasciicircum`.
- **Tilde** — `\~` produces an accent ($\tilde{n}$). Use `\textasciitilde`.

Examples

The cost is \$100, a 50% discount. Smith & Sons Ltd. The variable is named `my_var`. Use `\textbf{}` for bold.

The cost is `\$100`, a `50\%` discount.
Smith `\&` Sons Ltd.
The variable is named `\texttt{my\_var}`.
Use `\textbackslash textbf{\}` for bold.

Shortcut: Let verbatim Do It

Escaping by hand gets miserable fast when printing real code. `\verb` and the `verbatim` environment disable all processing, so nothing needs escaping:

`\textbf{100% & $50}` — printed exactly as typed.

This is why every code block in this document uses `verbatim`. Without it, a single backslash would have to be written as `\textbackslash` every single time.

In Math Mode

A few differ inside $\dots$:

- `\backslash` gives $\backslash$ (not `\textbackslash`)
- `\{` and `\}` still work for sets: $\{1, 2, 3\}$
- `\%` and `\&` behave the same as in text

11 References

11.1 Manual — `thebibliography`

Good for small, self-contained documents.

```
\begin{thebibliography}{9}    % {9} = width for 1 digit; use {99} for 10+
    \bibitem{meeus}
        Jean Meeus. \textit{Astronomical Algorithms}. Willmann-Bell Inc, 1991.
\end{thebibliography}
```

Cite with `\cite{meeus}` — e.g. this sentence cites the source [1].

11.2 BibTeX — `.bib` file

Better for papers and anything reused across documents. Create `refs.bib`:

```
@book{meeus1991,
  author   = {Jean Meeus},
  title    = {Astronomical Algorithms},
  publisher = {Willmann-Bell Inc},
  year     = {1991}
}
```

Then in the main file:

```
\bibliographystyle{plain}      % ACM-Reference-Format for acmart
\bibliography{refs}
```

Overleaf runs `bibtex` automatically. The `acmart` class expects `\bibliographystyle{ACM-Reference-Format}`.

12 Compile Checklist

1. **Compile twice** — cross-references (`\ref`, `\cite`) and the table of contents need a second pass. You will see `??` until then.
2. **Math mode** — `^` and `_` outside $\dots$ cause `Missing $ inserted`.
3. **Every row** in a `tabular` ends with `\\` before the next `\hline`.
4. **Empty &** for every cell a `\multirow` occupies from the row above.
5. `\label` after `\caption`, never before.
6. **Escape** `% & # _ $ { }` in text.

References

- [1] Jean Meeus. *Astronomical Algorithms*. Willmann-Bell Inc, 1991.